

Experiment No: 04

Name of the Experiment: Clipper and Clamper circuits.

Objective:

Study of Clipper and Clamper circuits.

Theory:

Clipper: Clippers remove signal voltage above and below a specified level. In the experiment no. 2, half wave rectifier can also be called as a clipper circuit. Because it clipped off the negative half cycle of the input signal.

A diode connected in series with the load can clipped off any half cycle of input depending on the orientation of the diode. (Figure 4.1) -

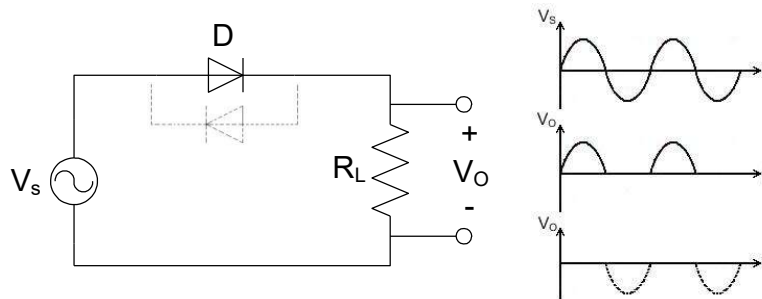


Figure 4.1: Simple Diode Clipper circuit

It is also possible to clip off a certain part of the input signal below a specified signal level by using a voltage source in reverse bias condition with the diode. If a battery of V volts is added to it, then for V_s above $(V+0.7)$ volts the diode becomes forward bias and turns ON. The load receives above this voltage Level.

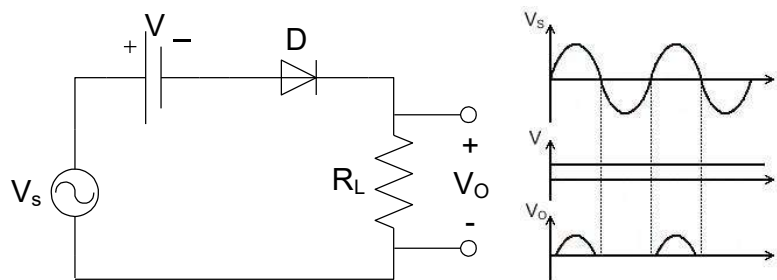


Figure 4.2: Clipper Circuit Using Bias Diode.

A diode connected in parallel with the load can clip off the input signal above 0.7 volts of one half cycle depending on the connection of the diode. Using two diodes in parallel in opposite direction both the half cycle can be limited to 0.7 volts.

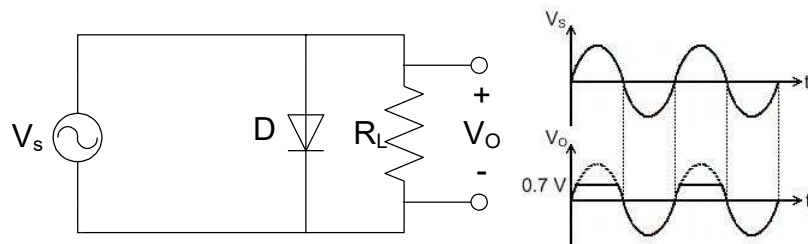


Figure 4.3: Parallel Clipper Circuit.

Using a biased diode it is possible to limit the output voltage to a specified level depending on the attached battery voltage. Either the half cycles or both of them can be clipped off above a specified level.

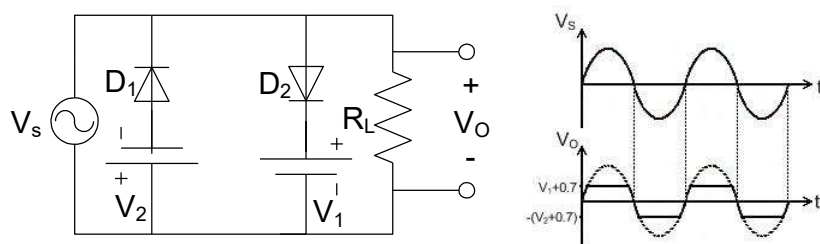


Figure 4.4: Biased Parallel Clipper Circuit.

In practical case for both the series and parallel clippers voltage source is not added. Required voltage levels are maintained by adding more semiconductor diode.

Clamper: A DC clamper circuit adds a DC voltage to the input signal. For instance, if the incoming signal varies from -10 volts to +10 volts, a positive DC clamper will produce an output that ideally swing from 0 volts to 20 volts and a negative clamper would produce an output between 0 volts to -20 volts.

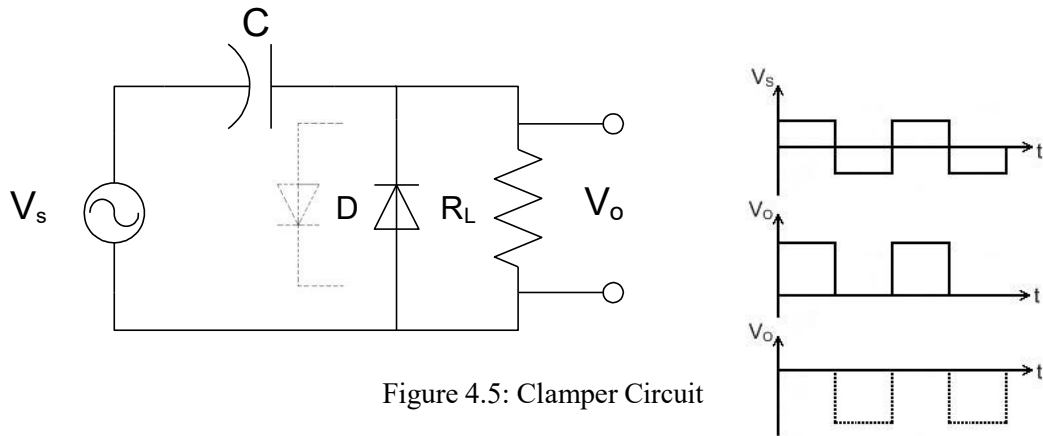


Figure 4.5: Clamper Circuit

Equipment and Components:

Serial no.	Component Details	Specification	Quantity
1.	p-n junction diode	1N4007	1 piece
2.	Resistor	100K Ω	1 piece
3.	Capacitor	0.1 μ F	1 piece
4.	Signal generator		1 unit
5.	Trainer Board		1 unit
6.	DC power Supply		1 unit
7.	Oscilloscope		1 unit
8.	Digital Multimeter		1 unit
9.	Chords and wire		as required

Experimental Setup:

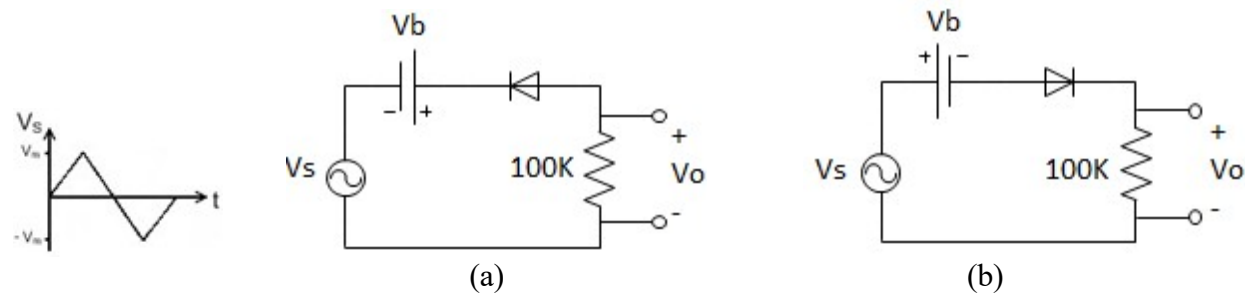


Figure 4.6: Series Clipper Circuit

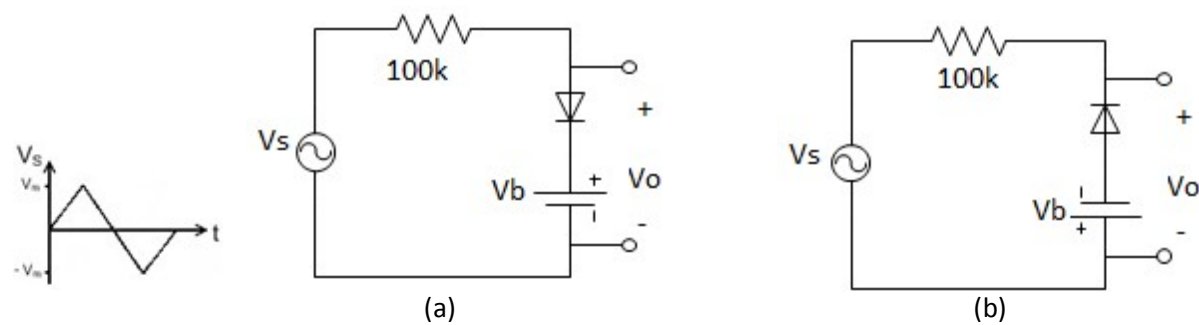


Figure 4.7: Parallel Clipper Circuits

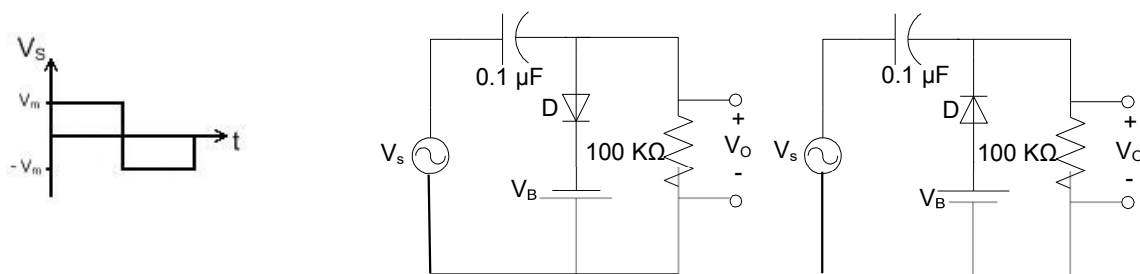


Figure 4.8: Clamper circuit

Procedure:

1. Connect the circuit as shown in the figure 4.6.
2. Using Signal generator, apply a 1kHz 10V_{p-p} sinusoidal voltage source input ($V_m = 5V$)
3. Fix V_b to 2.5V and In the same graph paper, Draw V_s and V_o .
4. Decrease the value of V_b from 2.5V to 0V, and observe the output wave shapes
5. Increase the value of V_b from 2.5V to 5V, and observe the output wave shapes
6. Repeat step 2-4 for figure 3.7 and figure 3.8
7. Record V_{max} and V_{min} for the output wave for the clamper circuit only for $V_b=2.5v$.

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Data Collection:

Signature of instructor:

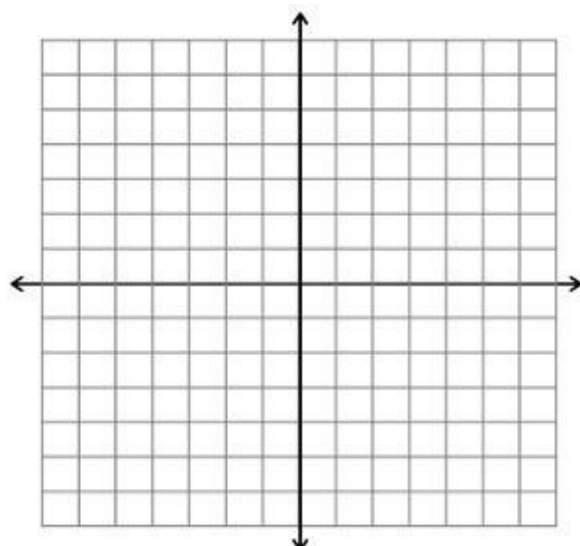
Experiment: 4,
Performed by Group# _____

Theoretical value: $R = \text{___ k}\Omega$

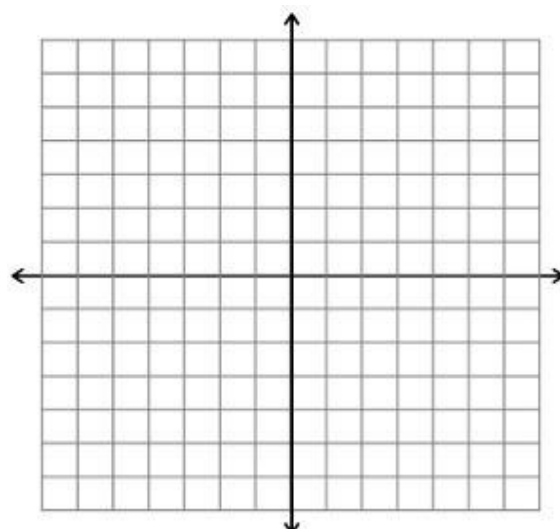
Measured value: $R = \text{___ k}\Omega$

$V_s = \text{___ V(p-p)}$.

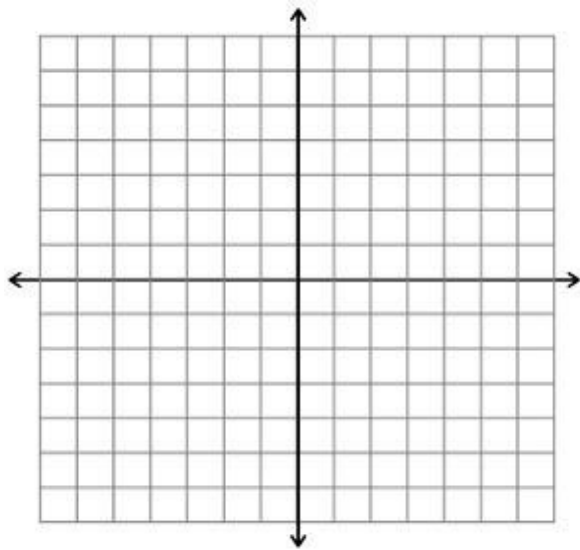
V_b (V)	V_o (p-p)					
	Fig 3.6		Fig 3.7		Fig 3.8	
	(a)	(b)	(a)	(b)	(a)	(b)
0						
1						
2						
2.5					$V_{\max} =$ $V_{\min} =$	$V_{\max} =$ $V_{\min} =$
3						
4						
5						



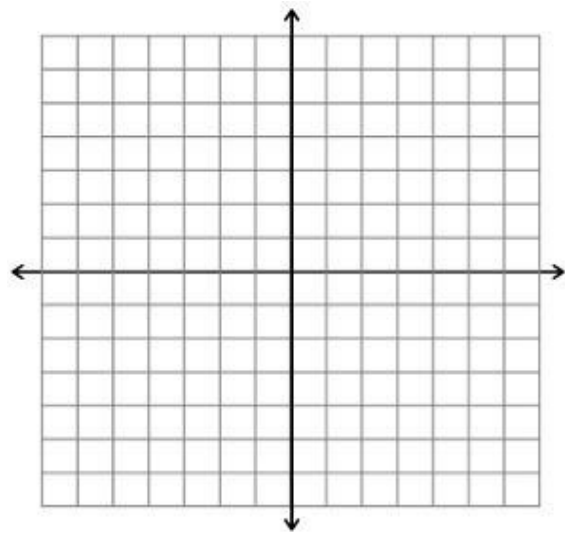
Input-output Fig 4.6 (a)



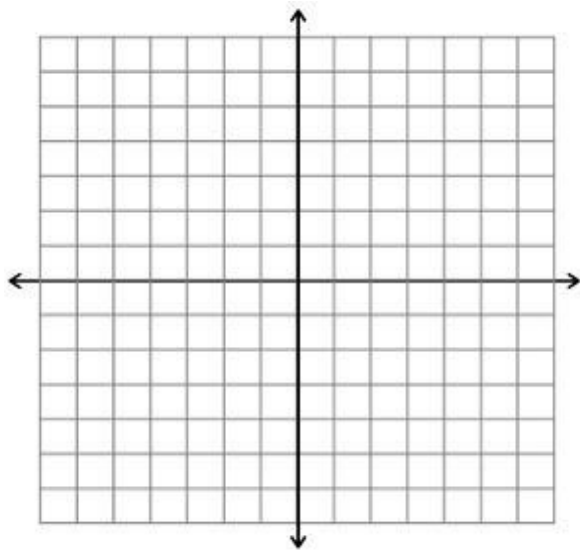
Input-output Fig 4.6 (b)



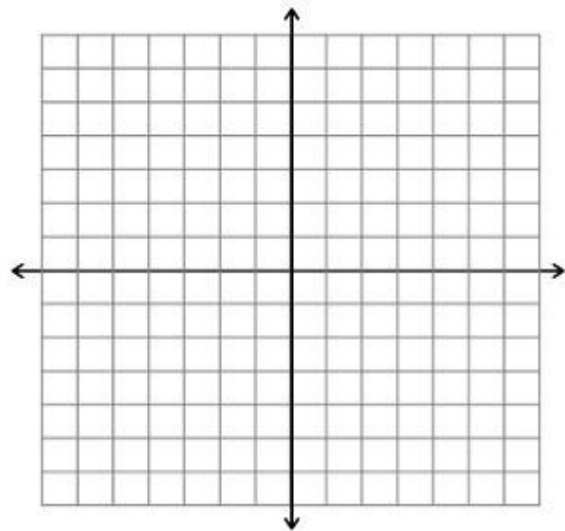
Input-output Fig 4.7 (a)



Input-output Fig 4.7 (b)



Input-output Fig 4.8 (a)



Input-output Fig 4.8 (b)

Report:

1. Using values from your data table, for all the circuit diagrams plot the input-output waveforms observed on the oscilloscope for $V_b=2.5V$.
2. For Fig 4.6(a &b), Fig4.7 (a & b) and Fig 4.8 (a & b) what change did you observe in the output voltage, In procedure-4? Explain the reason behind such a change.
3. For Fig 4.6(a &b), Fig4.7 (a & b) and Fig 4.8 (a & b) what change did you observe in the output voltage, In procedure-5? Explain the reason behind such a change.

Objectives:

Study of Clipper and Clamper circuit.

Theory:

A clipper circuit is a wave-shaping circuit that removes or clips a portion of an input signal without changing the remaining part of the waveform. It is commonly used in communications system, voltage protection circuits, and signal conditioning to limit the output voltage within a desired range. A clipper mainly uses a p-n junction diode, which conducts only when it is forward biased. For a silicon diode, the forward voltage drop is approximately 0.7 V. Therefore, the diode starts conducting when,

$V_D \gg 0.7 \text{ V}$, where V_D is the voltage across the diode.

Clipper circuits are of two types, series clipper and parallel clipper. In series clipper, the diode is connected in series with the load resistor. Depending on the diode orientation, either the positive or negative half-cycle of the input waveform is clipped. In a parallel clipper, the diode is connected across the load. When the input voltage exceeds the clipping level, the diode becomes forward biased and provides a low resistance path, limiting

the output voltage. By adding a DC source (V_B), the clipping level can be adjusted. For a biased clipper, the approximate clipping voltage is,

$V_{clip} \approx V_B + 0.7 \text{ V}$, where V_B is the applied bias voltage.

A clamper circuit is different from a clipper because it does not remove any part of the waveform. Instead, it shifts the entire waveform upward or downward by adding a DC component while maintaining almost the same peak-to-peak voltage. A clamper circuit consists of a diode, capacitor and resistor. During one half cycle of the input signal, the diode conducts ^{and} ~~the~~ changes the capacitor. During the opposite half cycle, the diode becomes reverse biased, and the capacitor acts as a temporary voltage source, shifting the entire waveform. Depending on the diode direction, the circuit acts as either a positive clamper or a negative clamper.

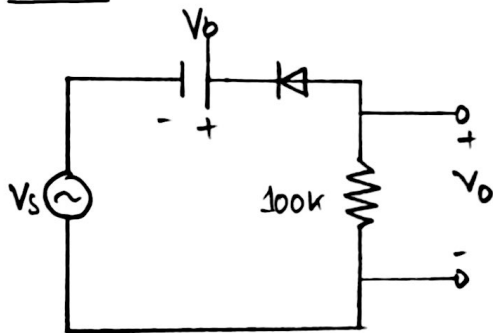
Apparatus List:

Serial No.	Component Details	Specification	Quantity
1	p-n junction node	1N4007	1 piece
2	Resistor	100 k Ω	1 piece
3	Capacitor	0.1 μ F	1 piece
4	Signal Generator		1 unit
5	Trainer Board		1 unit
6	DC power supply		1 unit
7	Oscilloscope		1 unit
8	Digital Multimeter		1 unit
9	Chords and Wire		as required.

Circuit Diagrams:

Series Clipper Circuit:

4.6 (a) :



Circuit Analysis:

For $0 \rightarrow T_{1/2}$

- $V_s < V_b$
- Diode = Reverse Biased
- $V_o = 0$

For, $T_{1/2} \rightarrow T$

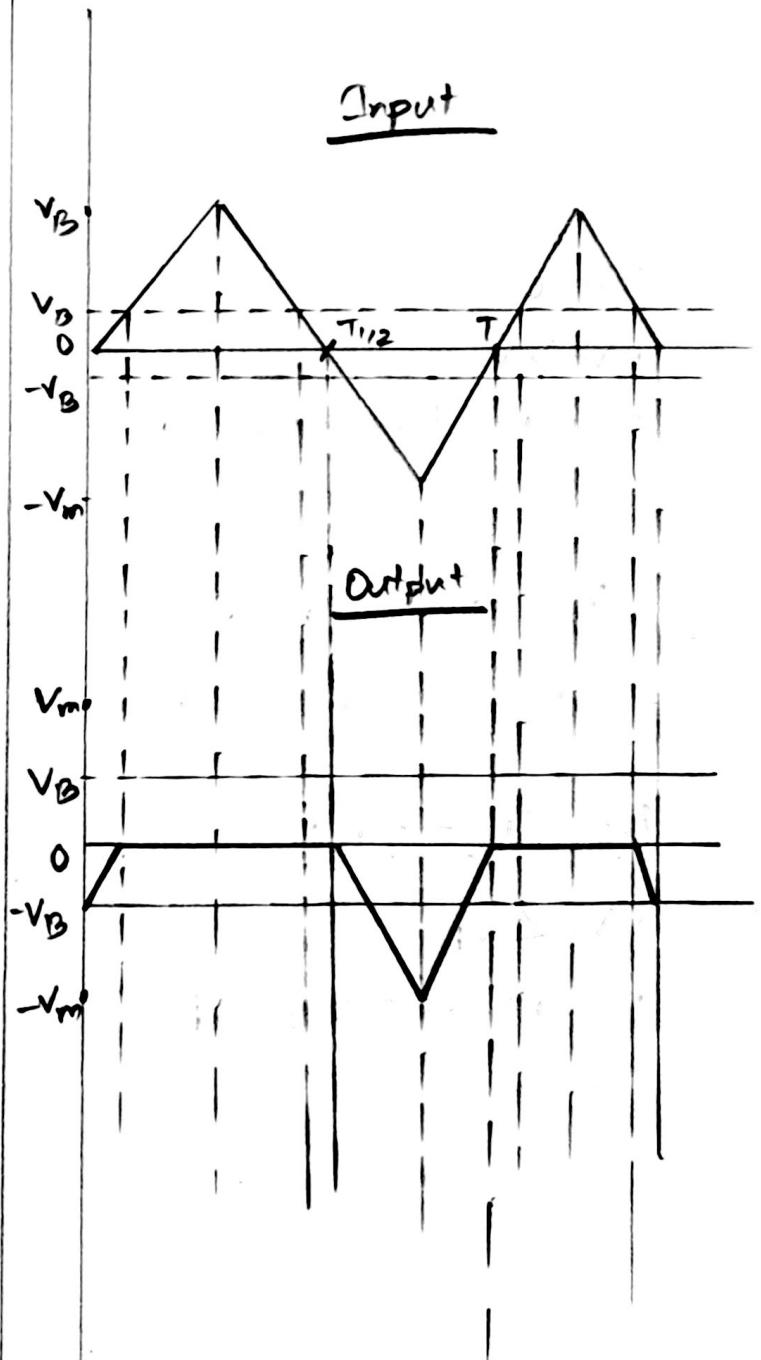
- $V_s > V_b$
- Diode = Forward

• Applying KVL,

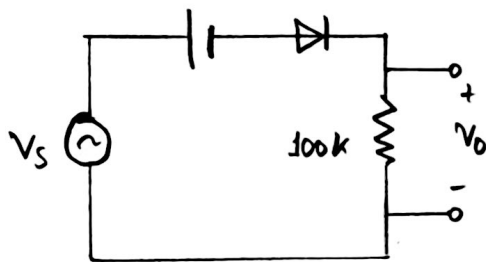
$$-V_s + V_D + V_b + V_o = 0$$

Since, $V_D = 0.7 \text{ V}$

$$\therefore V_o = \underline{V_s - V_b - 0.7 \text{ V}}$$



4.6(b)



Circuit Analysis:

For $0 \rightarrow T_{1/2}$

- $V_s > V_B$
- Diode = Forward Biased

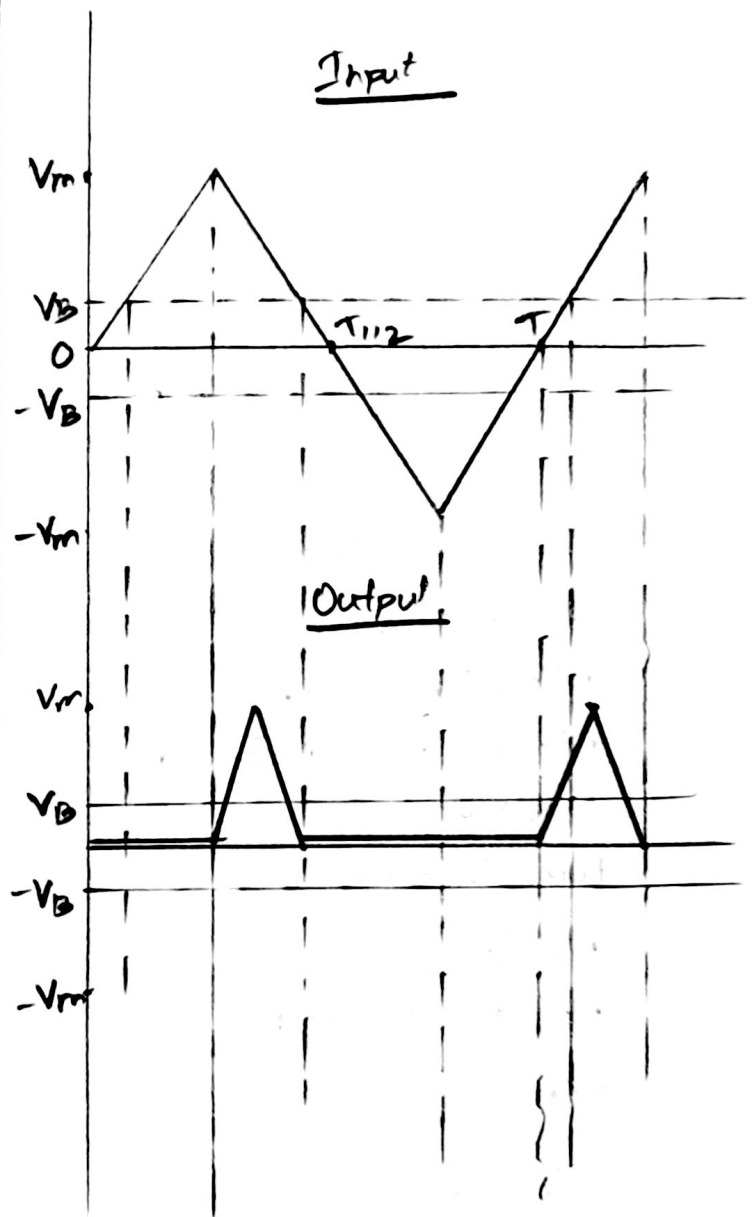
Applying KVL,

$$V_s - V_B - V_D + V_o = 0$$

$$\therefore V_o = -V_s + V_B + 0.7V$$

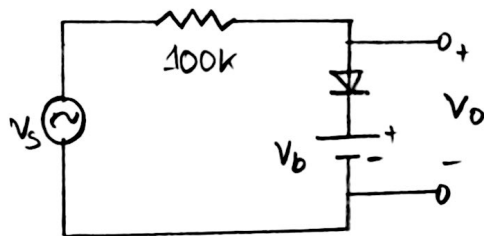
For $T_{1/2} \rightarrow T$:

- $V_s < V_B$
- Diode = Reverse Biased
- $V_o = 0$



Parallel Clippers Circuit:

4.7(a):



Circuit Analysis:

For $0 \rightarrow T_{1/2}$

- $V_s \rightarrow V_B$ $V_s < V_B$
- Diode = Reverse Bias
- $V_o = V_s$

For $T_{1/2} \rightarrow T$

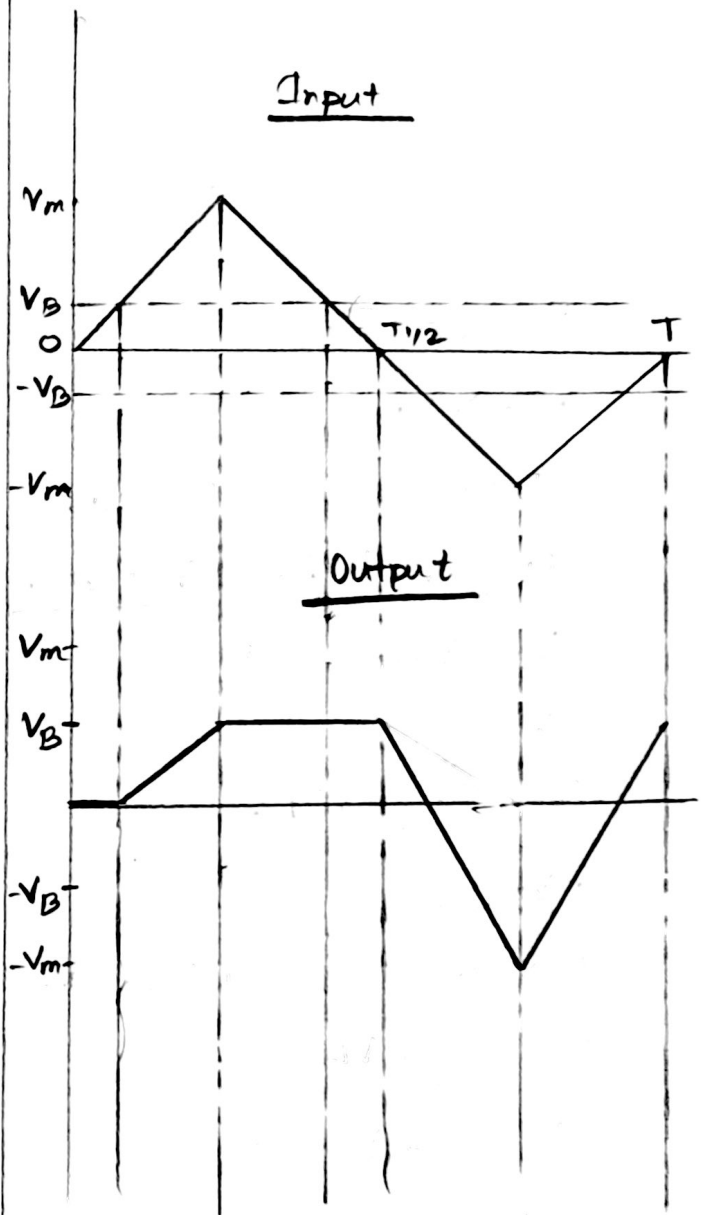
- $V_s > V_B$
- Diode = Forward Biased.

Applying KVL,

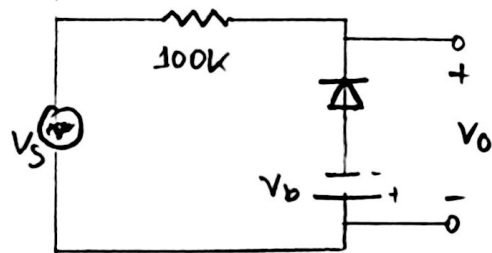
$$-V_o + V_D + V_B = 0$$

$$\Rightarrow V_o = V_B + V_D$$

$$\therefore V_o = \underline{V_B + 0.7V}$$



4.7(b)



For $0 \rightarrow T_{1/2}$:

- $V_S > V_B$
- Diode = Reverse Biased.

Applying KVL, $V_O = V_S$.

$$V_O + V_D + V_B = 0$$

$\Rightarrow$

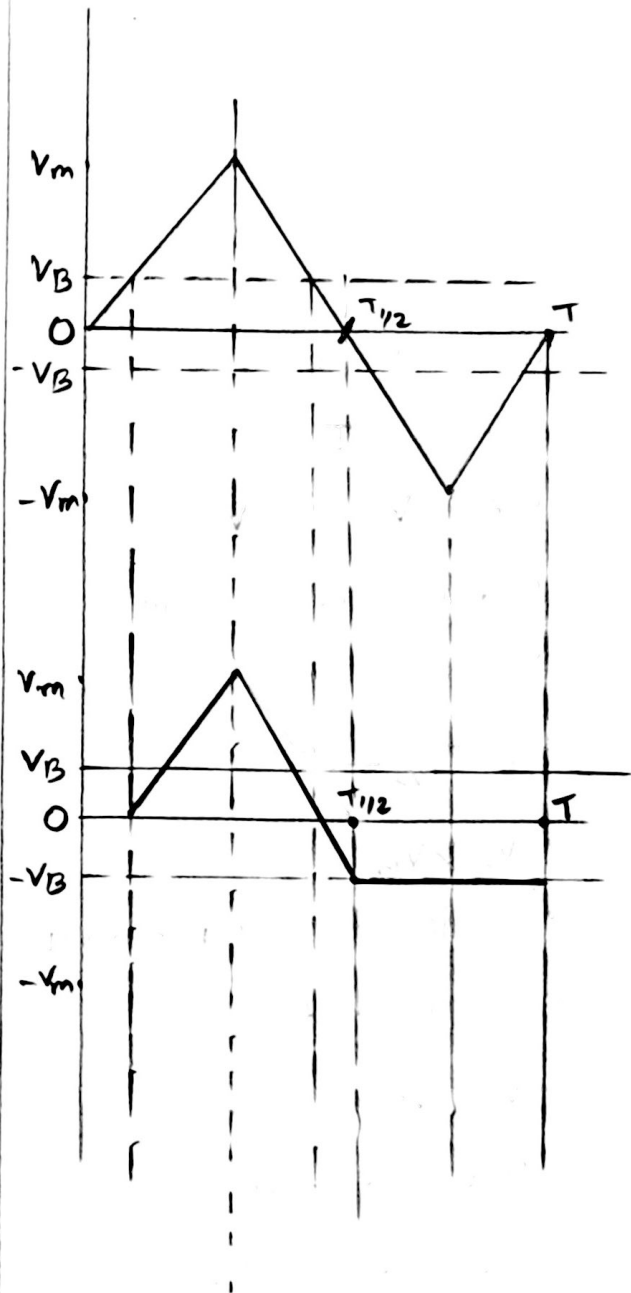
For, $T_{1/2} \rightarrow T$.

- $V_S < V_B$
- Diode = Forward Biased.

Applying KVL,

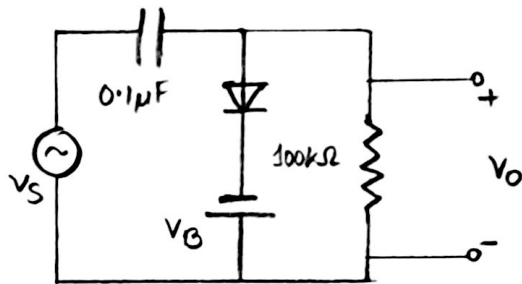
$$V_O + V_D + V_B = 0$$

$$\therefore V_O = -(V_B + 0.7 \text{ V})$$



Clamper Circuits:

4.8(a)



For $0 \rightarrow T_{1/2}$

- $V_s > V_B$
- Diode = Forward Bias
- Capacitor charges

Applying KVL,

$$-V_s + V_c + V_D + V_B = 0$$

$$\therefore V_c = V_s - V_B - 0.7 \text{ V}$$

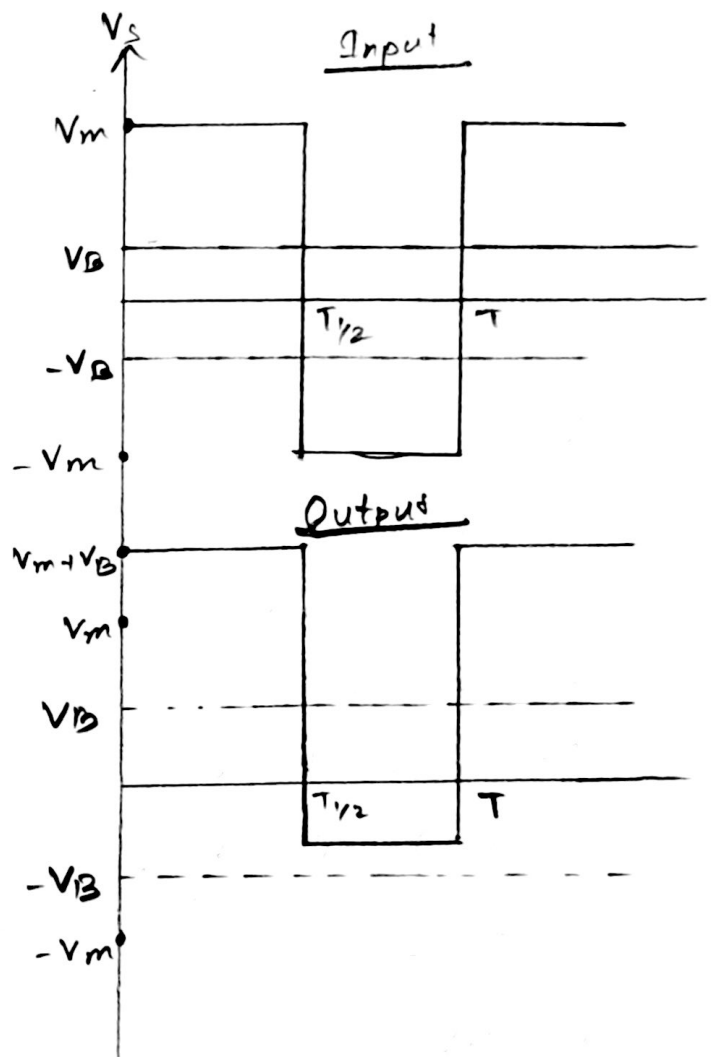
For $T_{1/2} \rightarrow T$,

- $V_s < V_B$
- Diode = Reverse Bias.
- Capacitor acts as a DC voltage source.

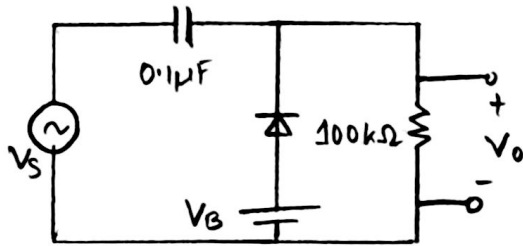
Applying KVL,

$$-V_s - V_c + V_o = 0$$

$$\therefore \underline{\underline{V_o = V_s + V_c}}$$



4.8 (b)



Circuit Analysis,

For $0 \rightarrow T_{1/2}$

- $V_s < V_B$
- Diode = Forward Bias

Applying KVL,

$$V_s + V_c - V_D - V_B = 0$$

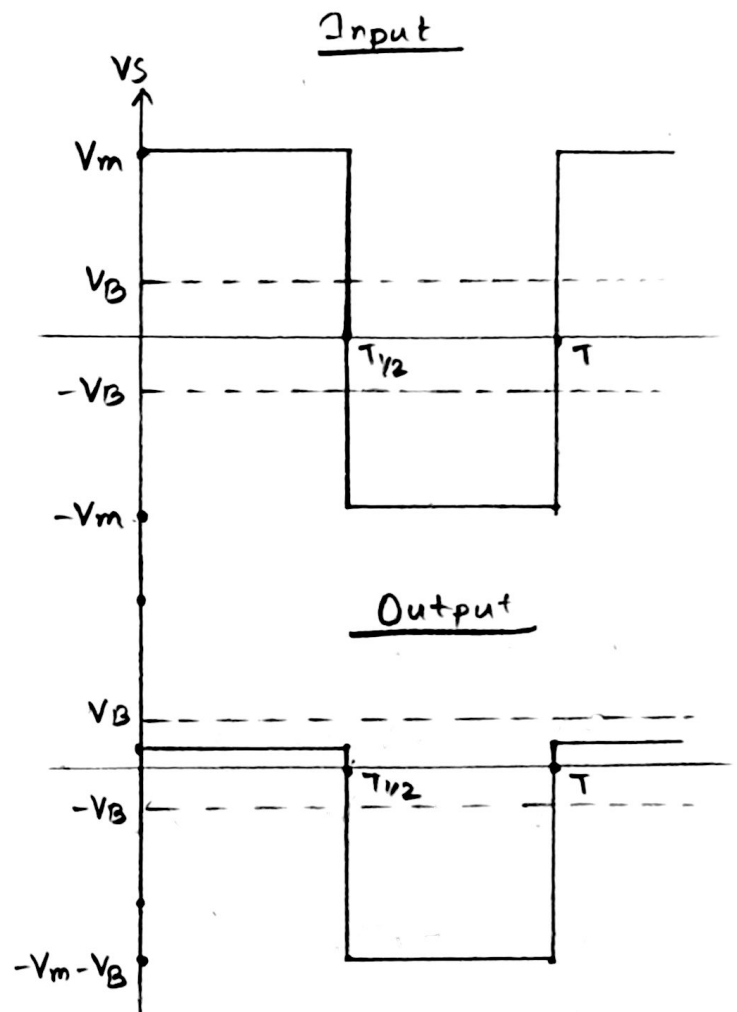
$$\therefore V_c = V_B + 0.7 - V_s$$

For $T_{1/2} \rightarrow T$

- $V_s > V_B$
- Diode = Reverse Bias
- Applying KVL,

$$-V_s + V_c + V_o = 0$$

$$\therefore V_o = \underline{V_s - V_c}$$



Data Table:

V_b (V)	V_o (p-p)					
	Fig 4.6		Fig 4.7		Fig 4.8	
	(a)	(b)	(a)	(b)	(a)	(b)
0	4.96 V	4.88 V	5.68 V	5.4 V	10.8 V	10.6 V
1	3.92 V	3.76 V	6.20 V	6.4 V	10.8 V	10.6 V
2	2.88 V	2.88 V	7.20 V	7.2 V	10.8 V	10.6 V
2.5	2.40 V	2.48 V	7.80 V	7.8 V	$V_{max} = -1.8 V$ $V_{min} = -12.6 V$	$V_{max} = 12.6 V$ $V_{min} = 1.8 V$
3	1.92 V	1.92 V	8.40 V	8.2 V	10.8 V	10.6 V
4	800 mV	800 mV	9.20 V	9.2 V	10.8 V	10.6 V
5	80 mV	80 mV	9.40 V	9.4 V	10.8 V	10.6 V

Result Analysis:

In this experiment we studied the working principle of series clipper, parallel clipper and clamper circuit by observing the input and output waveform of the oscilloscope. The results were close to the expected theory. For the series clipper circuit, it was seen that the output voltage changed as the bias voltage V_B was changed. For example, in Fig - 4.6(a), the output voltage decreased from 4.96 V at $V_B = 0$ to only 0.08 V at $V_B = 5V$,

because a large portion of the waveform was clipped. On the other hand, in fig. 4.6(b), the output voltage ^{also decreased} ~~increased~~ from 4.88 V to 0.08 V. This happened because a larger part of the input waveform was clipped as the bias voltage increased.

For the parallel clipper circuits, the opposite result was observed. In 4.7(a), the output voltage increased from 5.68 V at $V_B = 0$ V to 9.4 V at $V_B = 5$ V. In fig 4.7(b), the output voltage was also increased from 5.4 V to 9.4 V. The oscilloscope showed that only one side of the triangular waveform was clipped, which matched the expected behavior of the circuit.

For the clamper circuit, a square wave was used as the input signal. The shape and peak-to-peak voltage of the waveform remained almost the same, around 10.6 - 10.8 V, but the waveform shifted upward or downward depending on the diode connection. The maximum and minimum voltages changed because of the clamper circuit added a DC level without changing peak-to-peak voltage.

Question and Answers:

Solution 1: (Done in circuit diagram part).

Solution 2:

In procedure 4, the bias voltage (V_B) was decreased from 2.5 to 0 V. For the series clippers circuits, the output voltage increased because a smaller part of the input waveform was clipped. For example in fig 4.6 (a), the output voltage increased from 2.40 V to 4.96 V and in 4.6(b), it increased from 2.8 V to 4.88 V. For the parallel clippers circuits, the output voltage decreased from 7.8 V to 5.68 V in Fig 4.7(a), and in Fig 4.7(b) from 7.8 V to 5.4 V. In the clamper circuits, the output pk-to-pk voltage remained same almost ~~be~~ at about 10.6-10.8 V, but the waveform shifted because, the DC level changed. These changes happened because changing the bias voltage changes the diode conduction points, which changes the clipping level or the DC shift of the output waveform.

#Solution 3:

In procedure 5, the bias voltage (V_B) was increased from 2.5 V to 5 V. For the series clipper circuits, the output voltage decreased because a larger part of the input waveform was clipped. For the parallel clipper circuits, the output voltage increased because the clipping level shifted and more of the waveform appeared at the output. In the clippers circuits, the peak-to-peak voltage remained almost the same, but the waveform shifted upward or downward due to the change in the DC level. These changes occurred because increasing V_B changes the diode conduction point, which affects the output waveform.

Discussion:

From this experiment we learned how clippers and clamper circuits work and how a diode can clip or shift an input waveform. We also learned how changing the bias voltage (V_B) changes the output waveform in clipper circuit. By using the oscilloscope, we could ~~improve~~ compare the practical waveforms with the theoretical ones, which helped us understand the circuit operation more clearly.

The practical results were very close to the theoretical results. The clipper circuits clipped the required part of the triangular waveform, while the clamper circuits shifted the square waveform upward or downward without changing its peak-to-peak voltage. However, there were some small differences because of the diode's forward voltage drop, component tolerances and measurement errors.

During the experiment, we faced a problem with the oscilloscope because it was not showing the correct output waveform. At first we thought one of the components were faulty, so we replaced the diode, resistor and other components, but the problem remained still. Later we changed the breadboard, and the circuit started working.

properly. This showed that a faulty breadboard or poor internal connection can also affect the circuit and produce incorrect results.

Overall, this experiment improved our understanding of clippers and clamping circuits and also gave us practical experience in troubleshooting circuits, using the oscilloscope correctly, and analyze different waveforms. It also showed that careful circuit connections and proper testing are important for getting accurate results.

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Data Collection:

Experiment: 4,
Performed by Group# 11

Theoretical value: $R = 100 \text{ k}\Omega$

Measured value: $R = 96.7 \text{ k}\Omega$

$V_s = 10 \text{ V(p-p)}$.

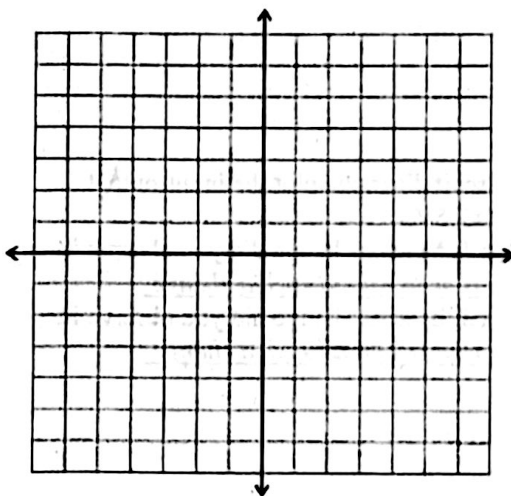
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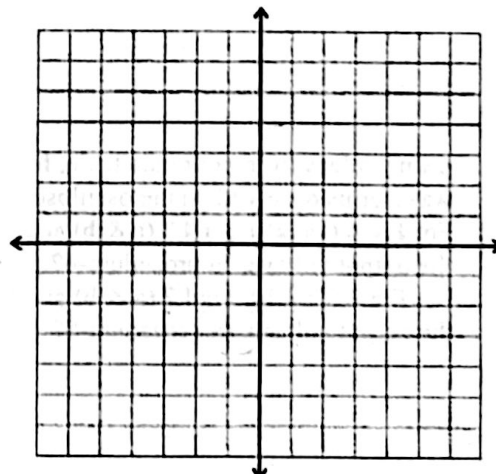
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V_b (V)	V_o (p-p)					
	Fig 3.6		Fig 3.7		Fig 3.8	
	(a)	(b)	(a)	(b)	(a)	(b)
0	4.98V	4.88V	5.68V	5.4V	10.8V	10.6V
1	3.92V	3.76V	6.20V	6.4V	10.8V	10.6V
2	2.88V	2.88V	7.2V	7.2V	10.8V	10.6V
2.5	2.40V	2.48V	7.8V	7.8	$V_{max} = -3.8V$ $V_{min} = -12.6V$	$V_{max} = 12.6V$ $V_{min} = 1.8V$
3	1.92V	1.92V	8.4V	8.2V	10.8V	10.6V
4	800mV	800mV	9.2V	9.2V	10.8	10.6V
5	80mV	80mV	9.4V	9.4V	10.8	10.6V



Input-output Fig 4.6 (a)



Input-output Fig 4.6 (b)